

QBITEL BRIDGE

INSURANCE & REINSURANCE SECURITY PLATFORM

Quantum-Safe Protection Across 30-50 Year Policy Lifecycles

30-50yr

Policy Data
Protection

\$80B+

Annual Fraud
Prevented

Solvency II

Ready
Compliant

78%

Autonomous
Threat Response

\$5.5B

Insurance Cyber Losses 2024

67%

Insurers on Legacy Mainframes

89%+

Protocol Discovery Accuracy

\$4.9M

Avg Insurance Breach Cost

AI-Powered | Quantum-Safe | Zero Disruption | Deployed in Hours

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EXECUTIVE SUMMARY

The Unique Quantum Risk of the Insurance Industry

The insurance industry faces a security crisis unlike any other sector: policy data created today must remain confidential for 30 to 50 years — precisely the timeframe in which quantum computers will break today's encryption. A life insurance policy issued in 2026 contains policyholder medical history, financial data, and beneficiary information that must stay protected until 2076. Today's harvest-now, decrypt-later attacks are already capturing that data.

QBITEL Bridge is the only AI-driven, quantum-safe protocol intelligence platform purpose-built for insurance workflows. It wraps every protocol — ACORD XML, X12 EDI (834/835/837), HL7, SWIFT, ISO 20022, TN3270e mainframe sessions, and proprietary policy administration traffic — in post-quantum cryptography without requiring re-architecture of existing systems.

\$5.5B

Insurance Cyber
Losses 2024

30-50yr

Policy Data
Lifecycle

67%

Insurers on
Legacy Mainframes

\$80B+

Annual Claims
Fraud Globally

- * *A life insurance policy issued today must protect policyholder medical and financial data for 60+ years. Quantum computers capable of breaking RSA-2048 will exist within 10-20 years. The harvest-now, decrypt-later threat is not theoretical — nation-state actors are capturing ciphertext today.*

THE INSURANCE INDUSTRY'S QUANTUM EXPOSURE

Harvest-Now, Decrypt-Later and the 30-50 Year Policy Lifecycle Risk

The Harvest-Now, Decrypt-Later Threat

No other industry has a data retention problem like insurance. A 25-year-old purchasing a whole life insurance policy in 2026 generates a policyholder record that must remain confidential for 60+ years. That record contains full medical underwriting history, financial assets and beneficiary designations, Social Security numbers and government IDs, actuarial risk classifications, and claims history across all lines of coverage.

Nation-state threat actors are systematically capturing encrypted insurance data today using the harvest-now, decrypt-later (HNDL) strategy. They archive ciphertext produced by RSA-2048 and ECC P-256, knowing that cryptographically relevant quantum computers will break these algorithms within 10-20 years. NIST finalised Post-Quantum Cryptography standards in August 2024 (FIPS 203/204/205). The transition has started.

Why Insurance is a Priority Target

Insurance companies hold the most complete picture of individual financial and health status of any private sector entity. The combination of health, financial, and family data in a single policyholder record is uniquely valuable for identity fraud, social engineering, and targeted extortion — and uniquely dangerous during the quantum transition.

Data Type	Retention	PQC Standard	Priority
Life/Annuity Policyholder Records	30-60 years	ML-KEM-1024 + ML-DSA-87	Critical
Health Insurance Subscriber Records	10-20 years	ML-KEM-768 + ML-DSA-65	High
P&C Policy Records	7-10 years	ML-KEM-768	High
Claims Payment Records	7 years	ML-KEM-512	Medium
Actuarial Model Data	Indefinite	ML-KEM-1024 + SLH-DSA	Critical
Reinsurance Treaty Data	20-30 years	ML-KEM-768 + ML-DSA-65	High

THREE CRITICAL THREATS FACING THE INSURANCE SECTOR

Long-Term Data | Legacy Systems | Claims and Reinsurance Fraud

Threat 1 > Long-Term Policyholder Data Exposure

Insurance policy records span 30-50 years. Data encrypted today with RSA or ECC will be decryptable by quantum computers within that window. Harvest-now, decrypt-later attacks are capturing policy data in transit right now — banking ciphertext for future decryption.

Every insurer with active whole life, annuity, long-term care, or disability policies is at risk. Health insurers holding multi-decade subscriber records and commercial insurers with D&O or professional liability policies are equally exposed.

NY DFS 500 requires encryption of NPI in transit and at rest. NAIC Model Law requires similar protections. Under Solvency II, EU-domiciled insurers must demonstrate current best practice protection — which now includes quantum-safe cryptography post-NIST PQC finalisation.

Threat 2 > Legacy Policy Administration System Vulnerabilities

67% of insurers run policy administration systems on mainframe platforms that predate modern cryptography — IBM System z, Unisys ClearPath, and COBOL-based systems using TN3270e terminal emulation. Communication between policy admin and downstream systems is often unencrypted.

Attack vectors include unencrypted TN3270e sessions, legacy ACORD XML over HTTP, cleartext X12 EDI 837 submissions from TPAs, and internal lateral movement that exploits the absence of per-session encryption on mainframe traffic.

Mainframe policy systems cannot be re-architected — they process millions of policies daily. Any security solution must operate as a transparent wrapper, not a replacement.

Threat 3 > Claims and Reinsurance Fraud

Insurance fraud costs the global industry \$80B+ annually. Synthetic identity attacks use composite identities combining real SSNs with fabricated credentials to obtain policies and file fraudulent claims. Staged accident rings are detectable as anomalous claims networks.

Reinsurance fraud includes falsified loss data in SWIFT settlement messages, business email compromise targeting high-value wire transfers for catastrophe loss settlements, catastrophe bond manipulation, and retrocession structure fraud.

QBITEL Bridge detects fraud at the wire level in real time — before claims are paid — providing a prevention layer that post-payment analytics cannot match.

7 CORE CAPABILITIES FOR INSURANCE

Protocol Security | Fraud Detection | Compliance | Mainframe Shield

Capability 1: Long-Term Policyholder Data Protection

The foundational challenge in insurance cryptography is the mismatch between algorithm lifespans

- Classify policyholder data by retention horizon — life, annuity, LTC, disability records identified for extended protection
- Apply CRYSTALS-Kyber (ML-KEM) key encapsulation for all policy data in transit, replacing RSA and ECC
- Wrap long-term storage with hybrid PQC+classical schemes ensuring quantum-safe forward secrecy
- Implement cryptographic agility — rotate PQC algorithms without disrupting policy data access
- Maintain per-policy key provenance logs suitable for regulatory audit and litigation hold

Capability 2: ACORD / X12 EDI Protocol Security

ACORD XML and X12 EDI carry policy issuance, claims submission, premium remittance, and reinsurance bordereau data.

- PQC wrapping of all ACORD XML message streams — inbound from agents/MGAs, outbound to reinsurers
- ML-DSA signature verification replacing SHA-1/MD5-based XML Digital Signatures on legacy integration points
- X12 EDI 834 (enrollment): ML-KEM PII protection, synthetic enrollment attack detection
- X12 EDI 835 (remittance): payment data integrity verification, payment redirection fraud detection
- X12 EDI 837 (claims): real-time integrity verification, upcoding and duplicate detection at the EDI layer
- HL7 v2.x and FHIR R4 PQC wrapping with PHI detection and HIPAA minimum necessary enforcement

Capability 3: Mainframe Policy System Shield

IBM System z and equivalent mainframes run COBOL policy engines communicating via TN3270e — typically unencrypted.

- Zero-Touch Deployment: transparent proxy with no mainframe code changes and no downtime
- TN3270e PQC Wrapping: every terminal session wrapped in ML-KEM with less than 0.8ms added latency
- Lateral Movement Prevention: session-level microsegmentation preventing perimeter breach escalation
- Privileged Session Monitoring: AI monitoring for privilege escalation, bulk exports, off-hours anomalies
- Native support: SNA/APPC, TN3270e, CICS transaction flows, JES job streams, IBM MQ policy messaging

Capability 4: Claims Fraud Detection and Prevention

AI-powered protocol analysis detects fraud at the wire level — before claims are processed — not after payment.

- Synthetic Identity Detection: cross-references 837 EDI identity signals against behavioral anomaly baselines
- Claims Network Analysis: detects organised fraud rings via coordinated submission pattern identification
- Medical Billing Integrity: real-time upcoding, unbundling, and phantom billing detection in 837 EDI
- Duplicate Claim Detection: cross-carrier detection using privacy-preserving hashed claim fingerprints
- SWIFT Message Integrity: real-time analysis of reinsurance settlement messages for manipulation
- Wire Fraud Prevention: ML-based BEC detection in SWIFT payment instruction chains for CAT settlements

Capability 5: Reinsurance Settlement Security

Catastrophe loss settlements may involve hundreds of millions in SWIFT wire transfers — a primary target for financial crime.

- SWIFT PQC: all SWIFT MT and MX (ISO 20022) reinsurance settlement messages protected with post-quantum cryptography
- ISO 20022 premium payment flows receive ML-KEM protection and ML-DSA message authentication
- FIX Protocol Security: ILS trading activity receives quantum-safe session encryption
- Treaty Data Protection: treaty terms and profit commissions receive long-term PQC matching treaty duration
- Settlement Integrity: cryptographic chaining creates tamper-evident audit trails for loss settlements

Capability 6: Solvency II / NY DFS 500 Compliance

Automated compliance evidence generation for every major insurance regulatory framework.

- Solvency II Pillar I: evidence of adequate data security controls for SCR model data protection
- Solvency II Pillar II/III: ORSA cybersecurity documentation and automated SFCR evidence generation
- NY DFS 500 Section 500.15: quantum-safe encryption of all NPI in transit and at rest
- NY DFS 500 Section 500.17: automated incident detection supporting 72-hour breach notification
- NAIC Model Law, HIPAA Security Rule, GDPR/CCPA, PCI-DSS v4.0, IFRS 17, SOC 2 Type II

Capability 7: Actuarial Data Integrity

Actuarial models represent decades of proprietary IP — pricing algorithms, mortality tables, CAT models, loss development factors.

- Model Data Encryption: all actuarial data encrypted with ML-KEM-1024 in transit between workstations and model servers
- Integrity Verification: ML-DSA signatures on model outputs create tamper-evident audit trails
- Access Anomaly Detection: AI monitoring identifies bulk export events, unusual queries, lateral movement
- Model Theft Prevention: protocol-layer DLP capabilities detect unauthorised large data transfers
- Reserve Data Protection: IBNR calculations receive cryptographic integrity verification

COMPLIANCE COVERAGE MATRIX

Automated Evidence Across All Major Insurance Regulatory Frameworks

Regulatory Framework	Jurisdiction	Key Requirement	QBITEL Coverage
Solvency II	EU / EEA	ICT security, ORSA controls, data protection	Full — automated evidence
NY DFS 500 (23 NYCRR 500)	New York	Encrypt NPI, MFA, incident response	Full — PQC + monitoring
NAIC Cybersecurity Model Law	24+ US States	Cybersecurity program, NPI protection	Full — protocol controls
HIPAA Security Rule	US (Health)	PHI encryption, access controls, audit	Full — PHI detection
GDPR	EU / EEA	Personal data protection, breach notification	Full — PII classification
CCPA / CPRA	California	Consumer data protection, security	Full — automated controls
PCI-DSS v4.0	Global	Premium payment encryption (Req 3 & 4)	Full — payment protection
IFRS 17	Global	Insurance contract data integrity	Full — audit trail chain
SOC 2 Type II	Global	Security trust service criteria	Full — continuous evidence
NIST CSF 2.0	US Federal	Cybersecurity framework alignment	Full — all 5 functions

INTEGRATION ECOSYSTEM

Native Connectors for Leading Insurance Technology Platforms

Platform Category	Supported Platforms	Integration Method
Policy Administration	Guidewire (PolicyCenter, ClaimCenter, BillingCenter), Duck Creek, Majesco, SAP Insurance, Sapiens	Native API + EDI wrapping
EDI & B2B Integration	IBM Sterling B2B Integrator, OpenText Trading Grid, Edifecs SpecBuilder/XEngine	Inline transparent proxy
Claims Management	Snapsheet AI Claims, Tractable, Mitchell International Reinsurance	API Layer + PQC wrapping
Reinsurance	SWIFT Network (MT/MX), FIX Protocol for ILS, RMS/Moodys CAT model data exchange	Protocol-layer overlay
Health Insurance	HL7 v2.x, FHIR R4, X12 clearinghouses, CMS integration points	PHI-aware wrapping

DEPLOYMENT TIMELINE

Zero-Downtime Deployment in Three Structured Phases

Phase 1: Protocol Discovery and Assessment — Days 1-3

- Passive network tap — no traffic disruption
- Automated discovery of ACORD XML, X12 EDI, HL7, SWIFT, mainframe protocols
- Data classification: all policyholder PII, PHI, NPI, and financial data streams identified
- Risk heat map: protocols ranked by sensitivity and quantum exposure
- Regulatory gap analysis: Solvency II, NY DFS 500, NAIC, HIPAA mapping

Deliverable: Insurance Protocol Security Assessment Report

Phase 2: PQC Overlay and Shield Deployment — Days 4-14

- HSM provisioning and PQC key generation (FIPS 140-3 Level 3)
- Bridge inline deployment — transparent insertion into protocol paths
- Mainframe Shield activation for TN3270e and legacy protocol streams
- ACORD XML and X12 EDI PQC wrapping activated by trading partner
- Fraud detection model activation calibrated to carrier claims mix
- Compliance monitoring dashboard activated

Deliverable: Go-Live Confirmation with Baseline Metrics

Phase 3: Optimisation and Compliance Reporting — Days 15-30

- Fraud detection model tuning based on carrier-specific claims patterns
- Regulatory evidence package generation (Solvency II SFCR, NY DFS 500, NAIC)
- Trading partner quantum-safe certificate distribution
- Actuarial team briefing and model protection validation
- Reinsurance counterparty coordination for quantum-safe settlement channels

Deliverable: First Compliance Evidence Package + 90-Day Roadmap

PERFORMANCE SPECIFICATIONS

Validated Enterprise Performance Benchmarks

Performance Metric	QBITEL Bridge Result
Protocol Discovery Accuracy	89%+ including undocumented legacy variants
PQC Encryption Overhead	Less than 1.2ms per session establishment
Claims EDI Processing Latency	Less than 0.8ms added on 837 streams
TN3270e Session Overhead	Less than 0.8ms per session
SWIFT Message Processing	Less than 1.5ms per MT/MX message
Autonomous Threat Response	78% resolved without human escalation
Fraud Detection Accuracy	Greater than 94% precision on synthetic identity patterns
Policy Transaction Throughput	2M+ transactions per day validated on mainframe environments
HSM Key Operations	FIPS 140-3 Level 3, 100,000+ operations per second
Availability SLA	99.99% with active-active high availability clustering
Deployment Time	4-6 hours for initial go-live; full deployment in 30 days

COMPETITIVE DIFFERENTIATION

Why Generic Security Solutions Fall Short for Insurance

Competitor Category	Limitation	QBITEL Bridge Advantage
Traditional Network Security	Layers 3-4 only — no ACORD/X12/TN3270e visibility	Layer 7 insurance protocol awareness and fraud detection
Cloud Encryption Services	Protects data at rest — not in-transit protocol streams	End-to-end, multi-layer PQC wrapping including mainframe and SWIFT
Fraud Analytics Platforms	Post-payment analytics — fraud detected after claim payment	Real-time pre-adjudication, inline detection at the EDI wire
Legacy PKI / Cert Management	No quantum-safe cryptography — RSA/ECC only	Only FIPS 203/204/205 PQC deployed today with crypto agnostic
Generic EDI Security	No ACORD awareness, no mainframe, no SWIFT leverage	Complete insurance protocol stack: EDI, ACORD, mainframe, SWIFT

CUSTOMER SCENARIOS

P&C Insurer | Life Insurer | Global Reinsurer

Scenario A > Regional P&C Insurer — Claims Fraud and EDI Security

Profile: Mid-size P&C insurer. 1.2M policies. \$800M annual claims volume. Duck Creek on-premises.
Challenge: 8% claims fraud losses. Three EDI trading partner incidents in 4 years. NY DFS 500 exam approaching.

Solution: Bridge deployed inline on Duck Creek EDI in 6 hours with zero downtime.
X12 837 real-time fraud detection identified 340+ suspicious claims in 30 days — synthetic identity ring across auto and workers compensation lines. NY DFS 500 Section 500.15 compliance achieved.

Results: \$4.2M in claims fraud prevented in 90 days. NY DFS 500 examination passed.

Scenario B > Large Life Insurer — Long-Term Data Protection and Mainframe Shield

Profile: National life/annuity insurer. 8M in-force policies. IBM System z. Solvency II EU subsidiary.
Challenge: Board quantum threat concern. No mainframe code changes accepted. APT campaign targeting.

Solution: Mainframe Shield as transparent proxy — TN3270e encrypted in less than 0.8ms.
6.3M policyholder records classified and protected with ML-KEM-1024. SWIFT reinsurance flows protected. Solvency II SFCR cybersecurity evidence package generated automatically.

Results: Solvency II regulator satisfied. Zero mainframe downtime. Three APT lateral movement attempts toward mainframe network segment detected and blocked.

Scenario C > Global Reinsurer — SWIFT Security and Catastrophe Settlement Integrity

Profile: Top-10 global reinsurer. \$45B assumed premiums. 200+ cedants. \$2.8B annual CAT SWIFT settlements.
Challenge: Two near-miss SWIFT payment fraud events in 18 months. PRA, EIOPA, BMA pressure.

Solution: SWIFT MT/MX PQC wrapping across all cedant settlement flows. Real-time amount and routing integrity verification. 200+ cedant treaty records protected with ML-KEM-1024.
CAT bond trigger data integrity via ML-DSA cryptographic chaining.

Results: Zero SWIFT fraud incidents post-deployment. \$180M CAT settlement protected.
Three regulatory jurisdiction examinations (PRA, EIOPA, BMA) passed with Bridge evidence packages.

NEXT STEPS & CONTACT

Start Your Insurance Quantum-Safe Journey

Step 01: Executive Briefing (30 minutes)

CRO/CISO-level session covering the quantum threat timeline for insurance, NY DFS 500 and Solvency II developments post-NIST PQC, and QBITEL Bridge capabilities. No technical prerequisites.

Step 02: Protocol Discovery Assessment (3 days, non-disruptive)

Passive tap assessment providing a full protocol inventory, data classification heat map, and regulatory compliance gap analysis. Deliverable: Insurance Protocol Security Assessment Report suitable for board presentation and regulatory examination preparation.

Step 03: Proof of Value Pilot (30 days, production scope)

Full Bridge deployment on a defined scope — single line of business, policy admin integration layer, or mainframe segment. Includes fraud detection activation, PQC wrapping, and compliance evidence generation. Full production conversion upon pilot success.

QBITEL Enterprise Insurance Practice
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QBITEL holds SOC 2 Type II certification. QBITEL Bridge is validated against NIST FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA). All insurance client engagements are conducted under mutual NDA. Response within 4 business hours for insurance sector enquiries.
